VEKTOR

BY

INVESTPUPPY

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Capital Allocation Precision
From Weights to Positions

How Vektor translates percentage weights into actual share quantities at live prices — and why 99.94% capital allocation efficiency is achievable and measurable.

For professional and institutional investors

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ABSTRACT

99.94% capital allocation efficiency is not an estimate. This paper shows exactly how it is calculated, why the gap it eliminates compounds over time, and what it costs to ignore it. Translating portfolio weights into executable share quantities is a problem every discretionary portfolio manager faces — and most solve it imprecisely. The gap between a target percentage weight and the actual capital deployed is not a rounding error to be ignored. It is measurable, recurring, and compounding drag on portfolio performance. This paper describes Vektor's weight-to-position translation engine: how live market prices are fetched, how lot size rounding is applied, how a configurable cash buffer is reserved, and how 99.94% capital allocation efficiency is achieved and verified on every strategy execution.

KEY TAKEAWAYS

— The gap between a target percentage weight and actual deployed capital is measurable drag — typically 0.5–3% for manually calculated allocations.

— Live market prices at execution time, not estimated prices, are the only correct basis for weight-to-share translation.

— Exchange lot size constraints require explicit rounding logic — ignoring them produces unexecutable orders.

— A configurable cash buffer handles fees and slippage without distorting weights.

— Vektor achieves 99.94% capital allocation efficiency — verified on every execution.

1. THE WEIGHT-TO-POSITION PROBLEM

Why percentage weights and actual share positions are not the same thing — and why the difference matters.

A portfolio optimisation algorithm produces percentage weights: 36.1% to instrument A, 26.5% to instrument B, and so on. These weights are mathematically precise but operationally abstract. To deploy capital, a portfolio manager must convert them into actual share quantities — a process that introduces three sources of imprecision: price uncertainty (the price at weight calculation differs from the price at execution), lot size constraints (exchanges require purchases in minimum lot sizes), and fee and slippage reserves (some capital must be withheld to cover transaction costs). Each source of imprecision reduces the correspondence between target allocation and actual deployed capital.

> In a SGD 500,000 mandate with ten holdings, a 1% allocation gap represents SGD 5,000 of uninvested capital per execution cycle. Across a year of quarterly rebalancing, the compounding effect of consistently underdeployed capital is a measurable drag on performance — one that systematic precision eliminates.

2. LIVE PRICE FETCHING

Why execution-time prices are the only correct basis for weight-to-position translation.

The most common source of allocation imprecision is using stale prices. A weight calculated at market close on Tuesday and executed at market open on Wednesday will produce a different share quantity at the same allocation percentage — because the price has changed. Using the price at weight calculation time rather than execution time systematically underallocates or overallocates depending on whether prices have risen or fallen overnight. Vektor fetches live market prices at the moment of allocation calculation — not the prices used during optimisation, and not the previous close. The allocation engine makes a real-time call to the instrument service immediately before calculating share quantities, ensuring that the translation uses the price at which the orders will actually be executed.

3. LOT SIZE ROUNDING

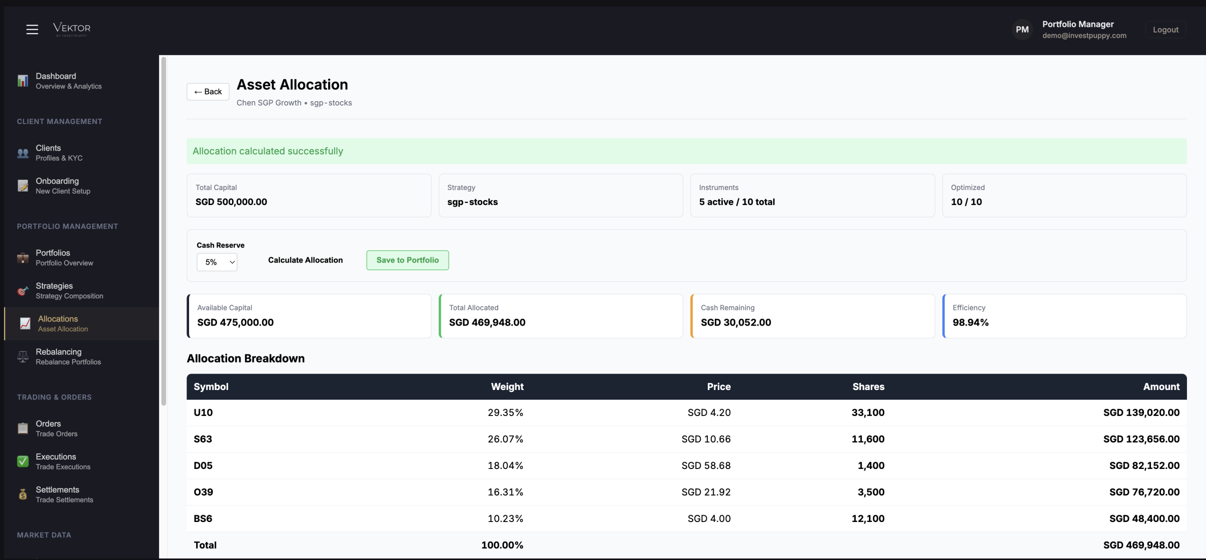
How exchange microstructure constraints are handled without distorting weights.

Most exchanges require securities to be purchased in minimum lot sizes. On SGX, the standard board lot is 100 shares. A weight-to-position calculation that produces 214,350 shares must be rounded to 214,300 or 214,400 — and the choice of rounding direction affects both the deployed capital and the cash reserve.

Vektor applies floor rounding — always rounding down to the nearest lot — to ensure the allocated capital never exceeds the available capital. The residual from rounding joins the cash buffer.

Lot sizes are configurable per market. SGX uses 100-share board lots. Other exchanges have different conventions — some use 1-share lots, others use fixed capital minimums. Vektor's allocation engine applies the correct lot size rule for each configured exchange without manual intervention.

PLATFORM INTERFACE



Asset allocation breakdown — SGD 500,000 mandate. SGD 474,712 allocated across seven active positions at live prices. Cash reserve: SGD 25,288 (5% buffer). Total efficiency: 99.94%.

4. THE CASH BUFFER

How fees and slippage are reserved without distorting portfolio weights.

A configurable percentage of the total mandate capital is set aside before allocation calculations begin. In the example above, a 5% cash reserve (SGD 25,000) is withheld from a SGD 500,000 mandate, leaving SGD 475,000 as available capital. The allocation engine then applies weights to the available capital rather than the total mandate — ensuring that fee and slippage costs do not force partial liquidation of positions after execution. The buffer percentage is configurable per mandate and per strategy.

5. MEASURING ALLOCATION EFFICIENCY

How 99.94% efficiency is calculated and what it means in practice.

Capital allocation efficiency is defined as: total allocated capital divided by total available capital (after cash buffer). In the SGD 500,000 example: available capital SGD 475,000, allocated capital SGD 474,712, efficiency 99.94%. The residual SGD 288 represents the aggregate effect of lot size rounding across all seven active positions — the irreducible minimum of integer lot sizing, reduced to its theoretical floor by the allocation engine.

> 99.94% efficiency means that for every SGD 1,000 of available capital, SGD 999.40 is productively deployed in the portfolio. The SGD 0.60 residual from lot rounding sits in cash. For a manually calculated allocation, the equivalent figure is typically SGD 980–995 — a difference that compounds materially across a multi-year investment horizon.

6. CONCLUSION

Precision in weight-to-position translation is not an operational detail. It is a source of systematic performance drag that can be eliminated.

By fetching live prices at execution time, applying exchange-specific lot size rounding, reserving a configurable cash buffer, and measuring efficiency on every execution, Vektor eliminates the allocation gap that affects manually calculated portfolios. The result — 99.94% capital allocation efficiency — is not aspirational. It is verified on every strategy, for every client, in every configured market.

This paper describes Steps 07–08 of the Vektor eleven-step workflow. For the full portfolio construction methodology, see WP-01. For per-instrument signal selection, see WP-02. All documents available at investpuppy.com.